

Electronic tasks

ID:734974

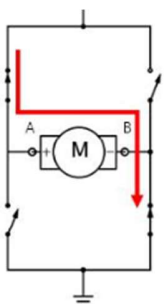
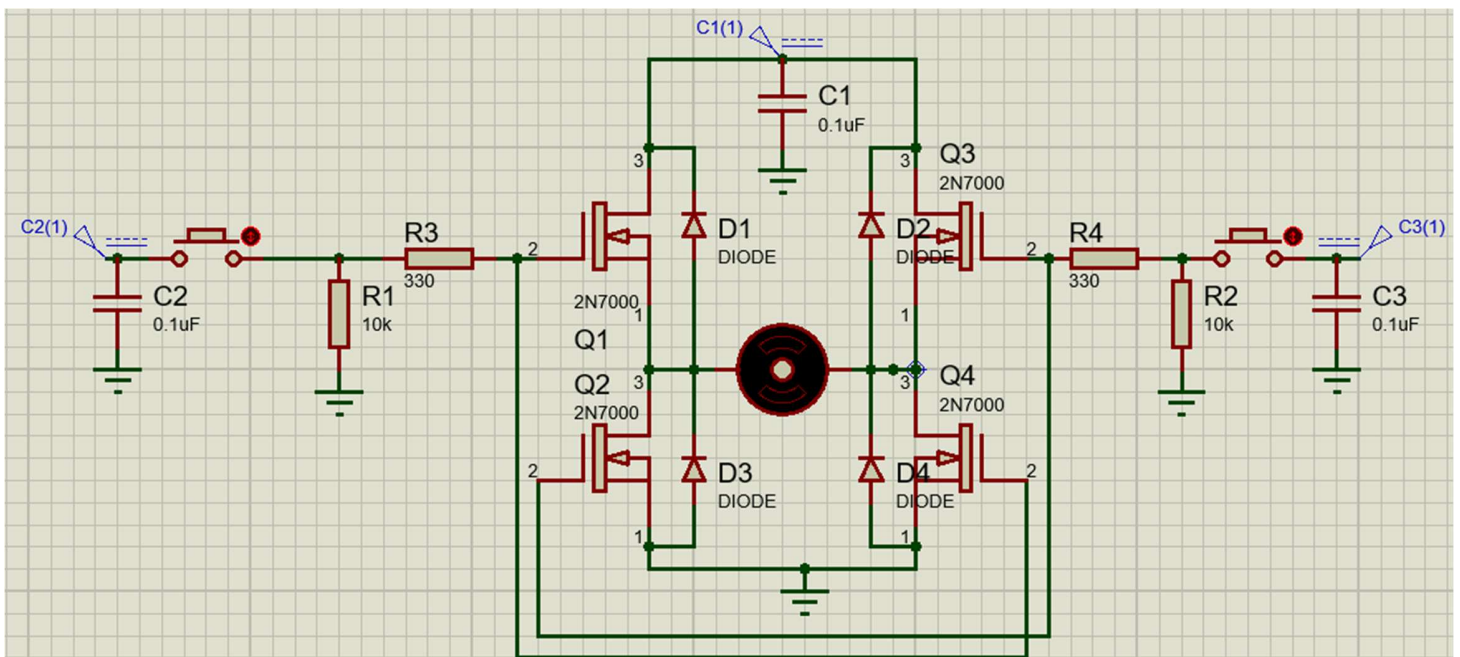
Task (1) “H Bridge”

An H-bridge is a circuit that allows a DC motor to rotate in both directions (clockwise and counterclockwise). The motor's rotation depends on the polarity of the voltage applied. This type of circuit is widely used in robotics, motor control, and automation systems.

Components used:

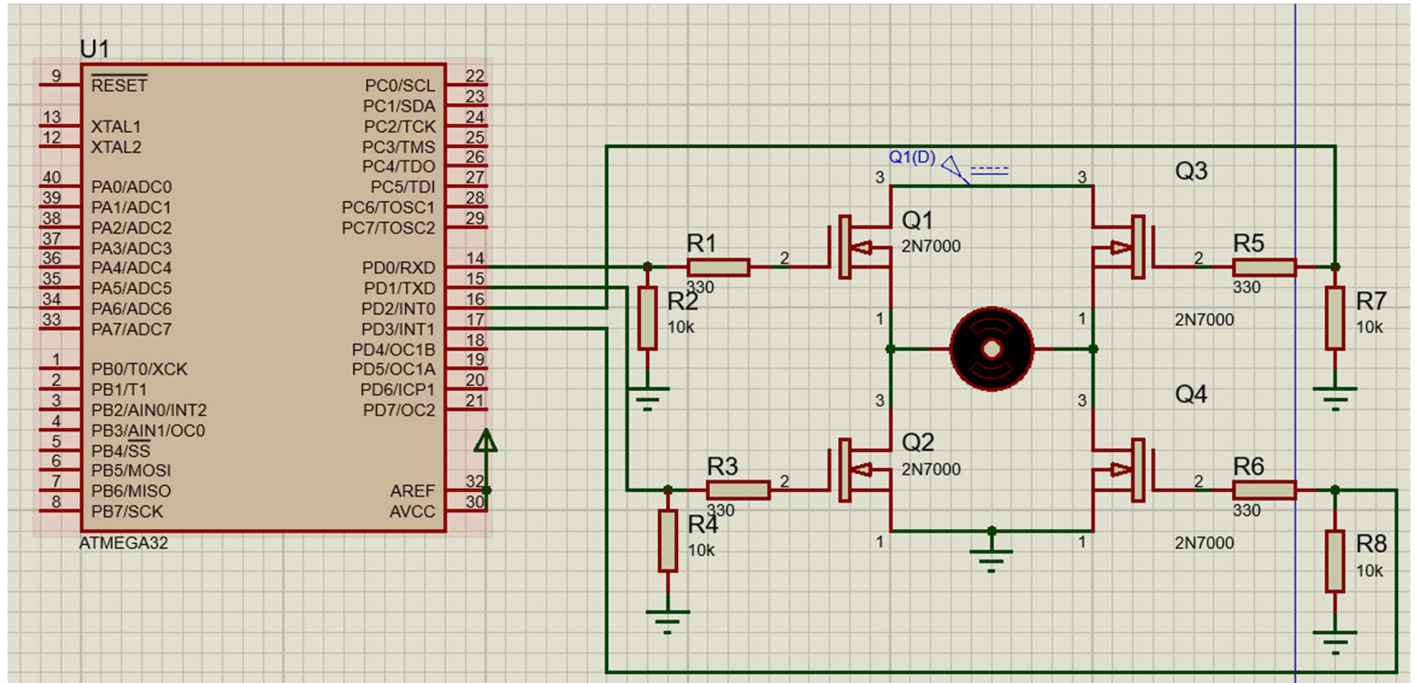
1. **N-Channel MOSFET:** used as switches that control the current flow.
2. **Push buttons:** used for manual control of motor direction (can be replaced with a microcontroller for automation).
3. **Pull down resistors:** make a return path that avoids damaging the microcontroller, and ensure MOSFET gates are at 0 V when no input is applied, preventing accidental switching.
4. **Capacitors:** used for decoupling to filter the signal noise, which improves the performance and stabilize the supply.
5. **Diodes:** protect the MOSFETs from the high voltage spikes.

Case 1 (using push buttons)



Case 2 “Using microcontroller (Atmega32):

A microcontroller can replace the push buttons, providing automatic control and to control the speed limit , we can use PWM with a duty cycle , higher duty cycle results in high speed while lower duty cycle reduces the speed .



Code :

```
#include <mega32a.h>
#include <delay.h>
void main(void)
{
    // Port D initialization
    DDRD=(0<<DDD7) | (0<<DDD6) | (0<<DDD5) | (0<<DDD4) | (1<<DDD3) | (1<<DDD2) | (1<<DDD1) | (1<<DDD0);
    // State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=0 Bit2=0 Bit1=0 Bit0=0
    PORTD=(0<<PORTD7) | (0<<PORTD6) | (0<<PORTD5) | (0<<PORTD4) | (0<<PORTD3) | (0<<PORTD2) |
    (0<<PORTD1) | (0<<PORTD0);
    while(1)
    {
        PORTD = (1<<PORTD0) | (1<<PORTD3); //ANTICLOCKWISE
        delay_ms(2000);

        PORTD = 0x00;          // STOP ALL THE GATES WITH ZERO CURRENT FLOW
        delay_ms(1000);

        PORTD = (1<<PORTD1) | (1<<PORTD2); // CLOCKWISE
        delay_ms(2000);

        PORTD = 0x00; // STOP AGAIN
        delay_ms(1000);
    }
}
```

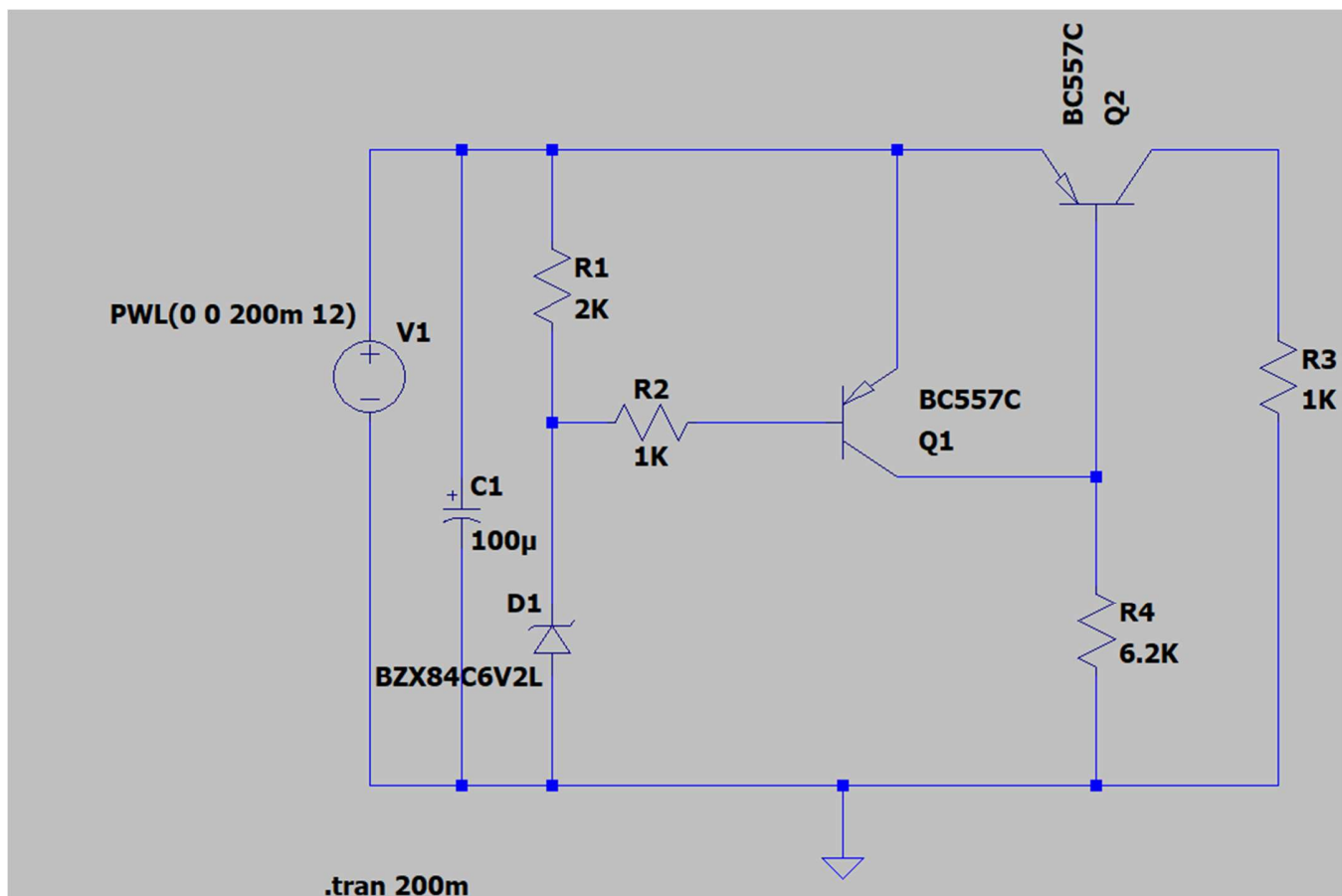
Tasks 2 “Overvoltage protection”

Case 1 (Using PNP):

Normally when the V_{in} doesn't exceed the threshold (V_z), the zener is not conducted so Q1 is not working and Q2 is working with a normal current flow to the load, once the V_{in} becomes greater than V_z the zener is conducted and Q1 is turned ON and Q2 is turned OFF which causes the current flow to be blocked forcing the output back to **zero** so the load is disconnected.

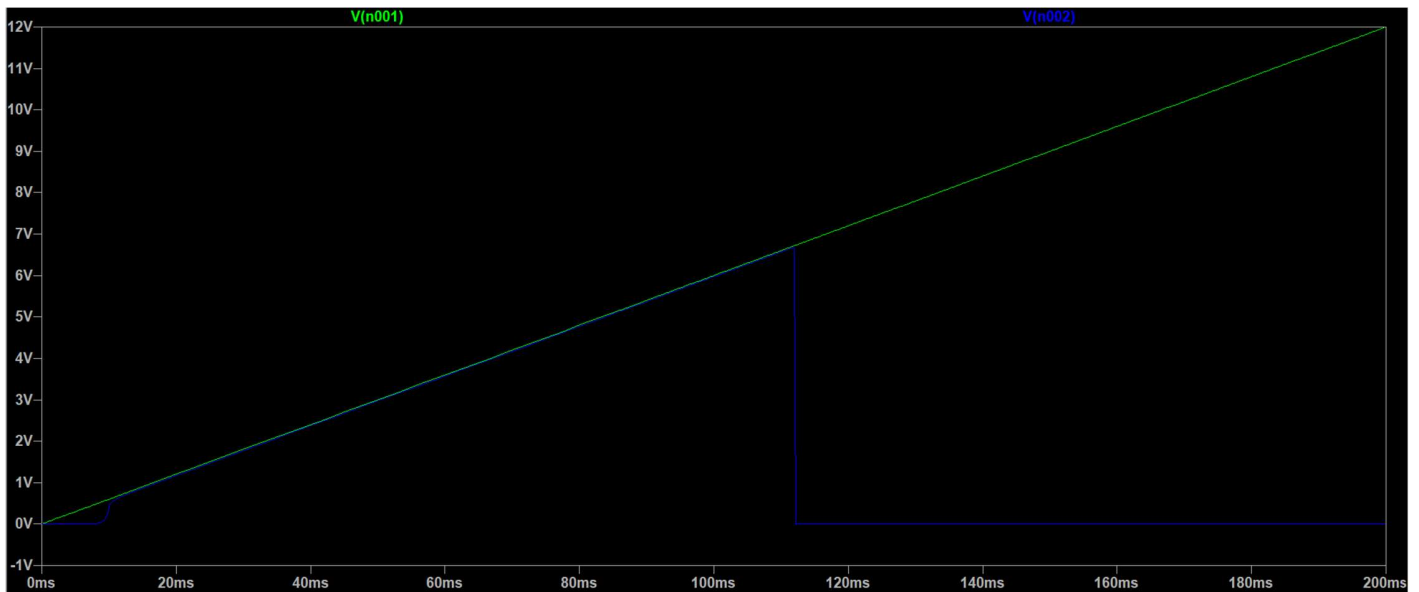
Components used :

1. 12V voltage source
2. Decoupling capacitor
3. 6.2V zener diode
4. Resistors (R3 is the load)
5. 2 PNP BJT



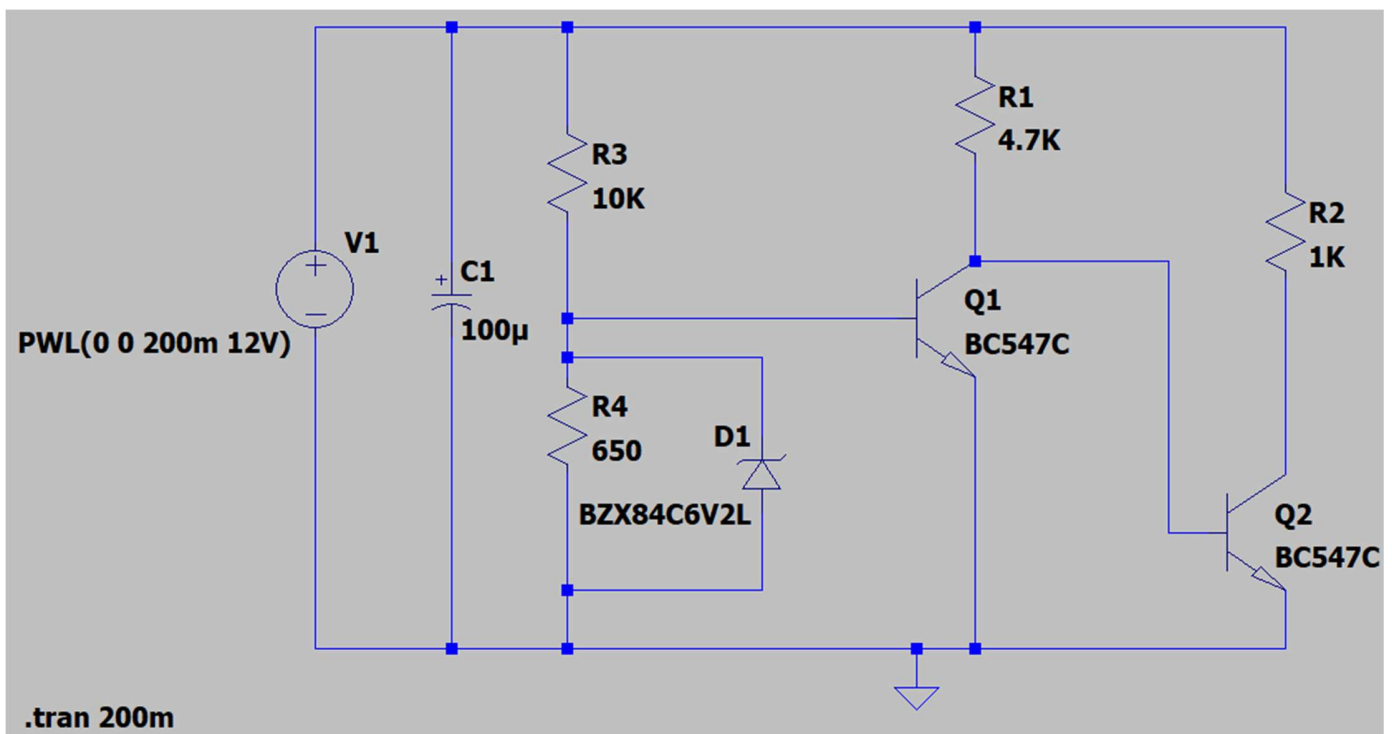
Explanation:

The output voltage remains zero till it exceeds 0.7V due to the silicon diode in the transistor and then it linearly rises till V_z . Once $V_{in} > V_z$ (Overvoltage) the current flow is blocked.



Case 2 (Using NPN)

NPN-based overvoltage protection circuit is designed to limit the output voltage to a safe level by shunting excess voltage to ground when the input exceeds a predefined threshold by using voltage divider and 2 transistors (Q1, Q2), Q1 is used as a switch that is triggered when the zener is conducted while q2 is used to divert the excess to ground.



Explanation:

If R4 is damaged, the zener diode will continue the circuit, so it's just a safety precaution.

